

BAB XVIII

PEMBUATAN LAPORAN

A. Tujuan Pembelajaran

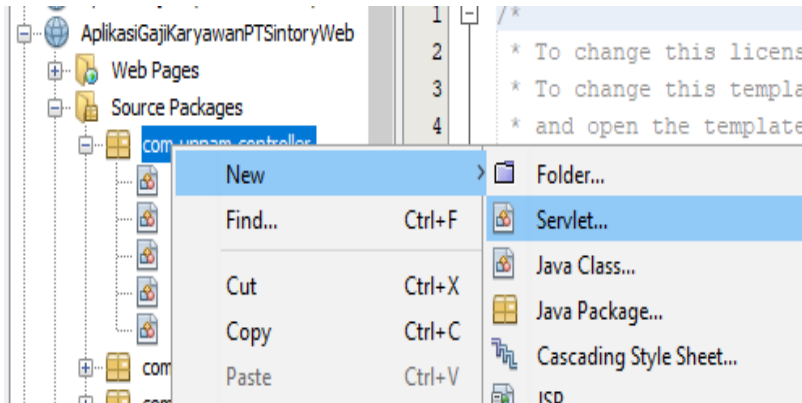
Pada pertemuan ini akan dijelaskan mengenai cara pembuatan laporan data pada database di aplikasi web menggunakan model MVC.

B. Materi

1. Pembuatan Laporan Pada Aplikasi Web

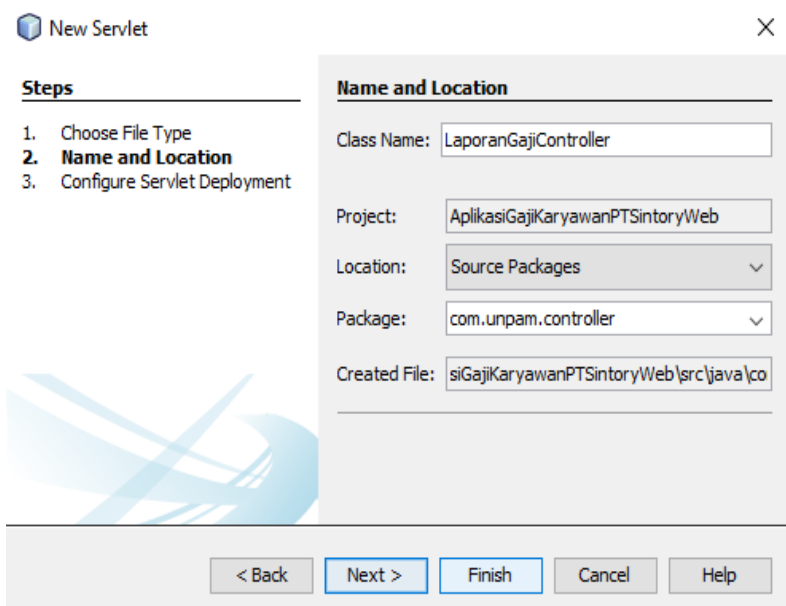
Langkah-langkah Pembuatan Laporan

Buatlah servlet LaporanGajiController pada package `com.unpam.controller`, klik kanan pada package `com.unpam.controller`, pilih New > Servlet.



Gambar 284. Membuat Servlet

Ubahlah Class Name menjadi LaporanGajiController, lalu klik Finish.



Gambar 285. Mengubah Class Name

Hapuslah source code berikut.

```
35 try (PrintWriter out = response.getWriter()) {  
36     /* TODO output your page here. You may use following sample code. */  
37     out.println("<!DOCTYPE html>");  
38     out.println("<html>");  
39     out.println("<head>");  
40     out.println("<title>Servlet LaporanGajiController</title>");  
41     out.println("</head>");  
42     out.println("<body>");  
43     out.println("<h1>Servlet LaporanGajiController at " + request.getContextPath() + "</h1>");  
44     out.println("</body>");  
45     out.println("</html>");  
46 }
```

Gambar 286. Menghapus Source Code

Silahkan tambahkan source code berikut didalam metode `processRequest` dibawah pernyataan `response.setContentType` pada `LaporanGajiController`.

```

41 String[][] formatTypeData = {"PDF (Portable Document Format)", "pdf", "application/pdf"},
42     {"XLSX (Microsoft Excel)", "xlsx", "application/vnd.openxmlformats-officedocument.spreadsheetml.sheet"},
43     {"XLS (Microsoft Excel 97-2003)", "xls", "application/vnd.ms-excel"},
44     {"DOCX (Microsoft Word)", "docx", "application/vnd.openxmlformats-officedocument.wordprocessingml.document"},
45     {"ODT (OpenDocument Text)", "odt", "application/vnd.oasis.opendocument.text"},
46     {"RTF (Rich Text Format)", "rtf", "text/rtf"};
47
48 HttpSession session = request.getSession(true);
49 String userName = "";
50
51 String tombol = request.getParameter("tombol");
52 String opsi = request.getParameter("opsi");
53 String ktp = request.getParameter("ktp");
54 String ruang = request.getParameter("ruang");
55 String formatType = request.getParameter("formatType");
56
57 if (tombol==null) tombol="";
58 if (ktp==null) ktp="";
59 if (opsi==null) opsi="";
60 if (ruang==null) ruang="0";
61
62 String keterangan = "<br>";
63 int noType = 0;
64
65 for (int i=0; i<formatTypeData.length; i++){
66     if (formatTypeData[i][0].equals(formatType)){
67         noType = i;
68     }
69 }
70
71 try {

```

```

71 try {
72     userName = session.getAttribute("userName").toString();
73 } catch (Exception ex){}
74
75 if (!(userName == null) || userName.equals("")){
76     boolean opsiSelected = false;
77
78     if (tombol.equals("Cetak")){
79         Gaji gaji = new Gaji();
80
81         int ruangDipilih = 0;
82         try {
83             ruangDipilih = Integer.parseInt(ruang);
84         } catch (NumberFormatException ex){}
85
86         if (gaji.cetakLaporan(opsi, ktp, ruangDipilih, formatTypeData[noType][1], getServletConfig().getServletContext().getRealPath("reports/GajiReport.jpg"),
87             byte[] pdfasbytes = gaji.getPdfasbytes();
88             try (OutputStream outputStream = response.getOutputStream()) {
89                 response.setHeader("Content-Disposition", "inline; filename=GajiReport."+formatTypeData[noType][1]);
90                 response.setContentType(formatTypeData[noType][2]);
91
92                 response.setContentLength(pdfasbytes.length);
93                 outputStream.write(pdfasbytes, 0, pdfasbytes.length);
94
95                 outputStream.flush();
96                 outputStream.close();
97             }
98         } else {
99             keterangan += gaji.getPesan();
100         }
101     }

```

```

101 }
102
103 String konten = "<h2>Mencetak Gaji</h2>";
104 konten += "<form action='LaporanGajiController' method='post'>";
105 konten += "<table>";
106 konten += "<tr>";
107 if (opsi.equalsIgnoreCase("KTP")){
108     konten += "<td align='right'><input type='radio' checked name='opsi' value='KTP'></td>";
109     opsiSelected = true;
110 } else {
111     konten += "<td align='right'><input type='radio' name='opsi' value='KTP'></td>";
112 }
113 konten += "<td align='left'>KTP</td>";
114 konten += "<td align='left'><input type='text' value='"+ktp+"' name='ktp' maxlength='15' size='15'></td>";
115 konten += "</tr>";
116
117 konten += "<tr>";
118 if (opsi.equals("ruang")){
119     konten += "<td align='right'><input type='radio' checked name='opsi' value='ruang'></td>";
120     opsiSelected = true;
121 } else {
122     konten += "<td align='right'><input type='radio' name='opsi' value='ruang'></td>";
123 }
124 konten += "<td align='left'>Ruang</td>";
125 konten += "<td align='left'>";
126 konten += "<select name='ruang'>";
127 konten += "<option selected value=0>Semua</option>";
128 for (int i=1; i<=14; i++) {
129     if (i==Integer.parseInt(ruang)){
130         konten += "<option selected value="+i+">"+i+"</option>";
131     } else {

```

```

131     } else {
132         konten += "<option value="+i+">"+i+"</option>";
133     }
134 }
135 konten += "</select>";
136 konten += "</td>";
137 konten += "</tr>";
138
139 konten += "<tr>";
140 if (!opsiSelected){
141     konten += "<td align='right'><input type='radio' checked name='opsi' value='Semua'></td>";
142 } else {
143     konten += "<td align='right'><input type='radio' name='opsi' value='Semua'></td>";
144 }
145 konten += "<td align='left'>Semua</td>";
146 konten += "<td><br></td>";
147 konten += "</tr>";
148
149 konten += "<tr>";
150 konten += "<td colspan='3'><br></td>";
151 konten += "</tr>";
152
153 konten += "<tr>";
154 konten += "<td>Format Laporan</td>";
155 konten += "<td colspan=2>";
156 konten += "<select name='formatType'>";
157 for (String[] formatLaporan : formatTypeData){
158     if (formatLaporan[0].equals(formatType)){
159         konten += "<option selected value='"+formatLaporan[0]+"'+>"+formatLaporan[0]+"</option>";
160     } else {
161         konten += "<option value='"+formatLaporan[0]+"'+>"+formatLaporan[0]+"</option>";

```

```

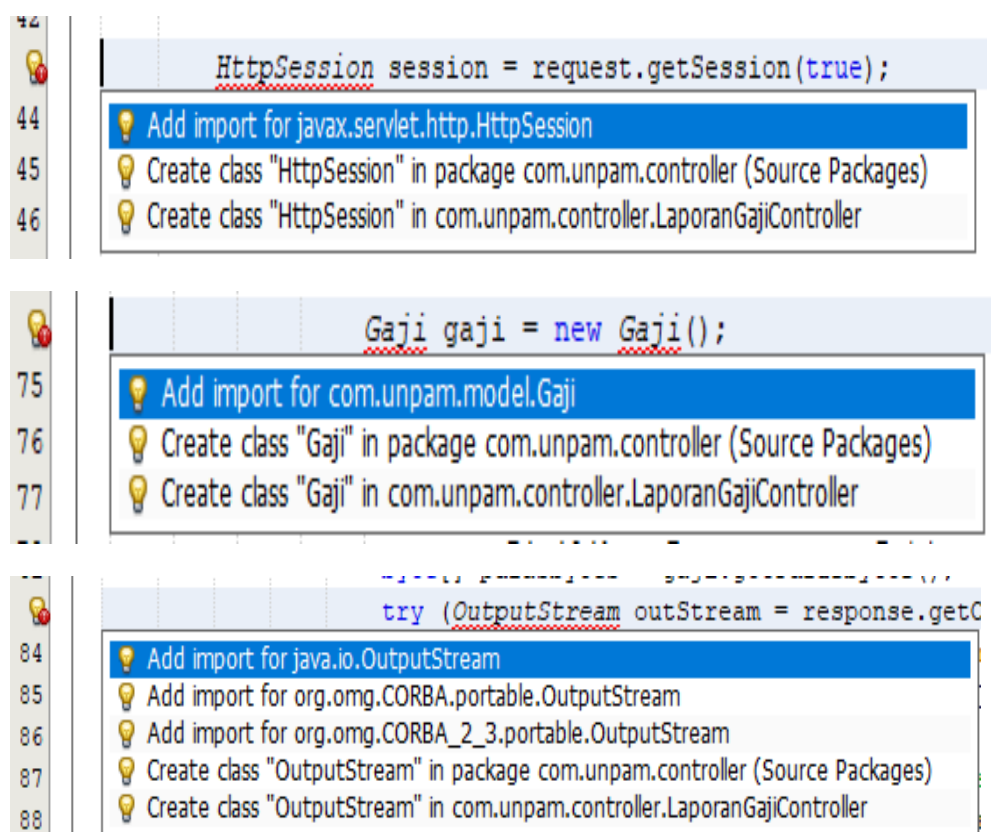
160         } else {
161             konten += "<option value='"+formatLaporan[0]+"'+formatLaporan[0]></option>";
162         }
163     }
164     konten += "</select>";
165     konten += "</td>";
166     konten += "</tr>";
167
168     konten += "<tr>";
169     konten += "<td colspan='3'><b>" + keterangan.replaceAll("\n", "<br>").replaceAll(";", ", ") + "</b></td>";
170     konten += "</tr>";
171
172     konten += "<tr>";
173     konten += "<td colspan='3' align='center'><input type='submit' name='tombol' value='Cetak' style='width: 100px'></td>";
174     konten += "</tr>";
175
176     konten += "</table>";
177     konten += "</form>";
178
179     new MainForm().tampilkan(konten, request, response);
180 } else {
181     response.sendRedirect(".");
182 }
183
184
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```

HttpServlet methods. Click on the + sign on the left to edit the code.

Gambar 287. Menambahkan Source Code

Klik semua lampu indikator kesalahan (lampu merah), dan pilih dengan Add Import....



Gambar 288. Memperbaiki Indikator Kesalahan

Jika pada `try (OutputStream....` masih ada yang salah, maka pindahkan `OutputStream` diatas `try` seperti berikut:

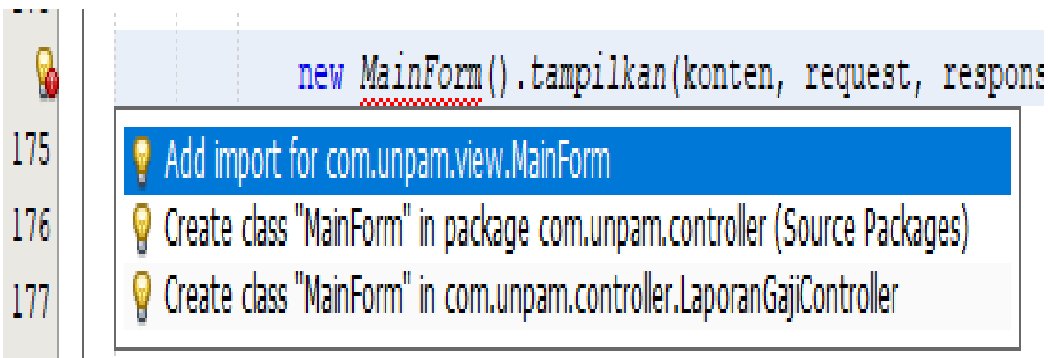
```

84 try (OutputStream outputStream = response.getOutputStream()) {
85     response.setHeader("Content-Disposition", "inline; filename=GajiReport."+formatTypeData[noType][1]);
86     response.setContentType(formatTypeData[noType][2]);
87
88     response.setContentLength(pdfasbytes.length);
89     outputStream.write(pdfasbytes, 0, pdfasbytes.length);
90
91     outputStream.flush();
92     outputStream.close();
93 }

```

Gambar 289. Menambahkan Source Code

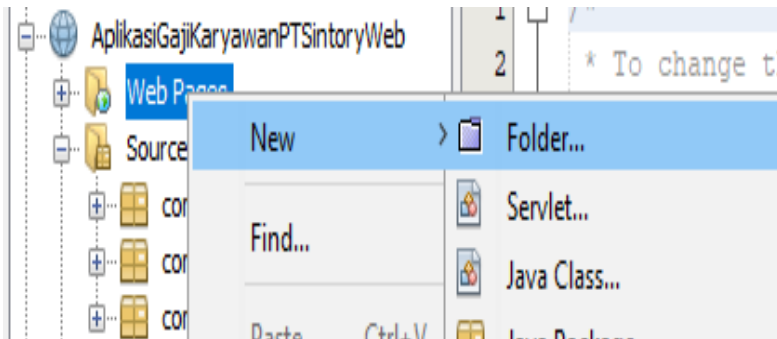
Klik lampu indikator kesalahan, pilih Add Import....



Gambar 290. Memperbaiki Indikator Kesalahan

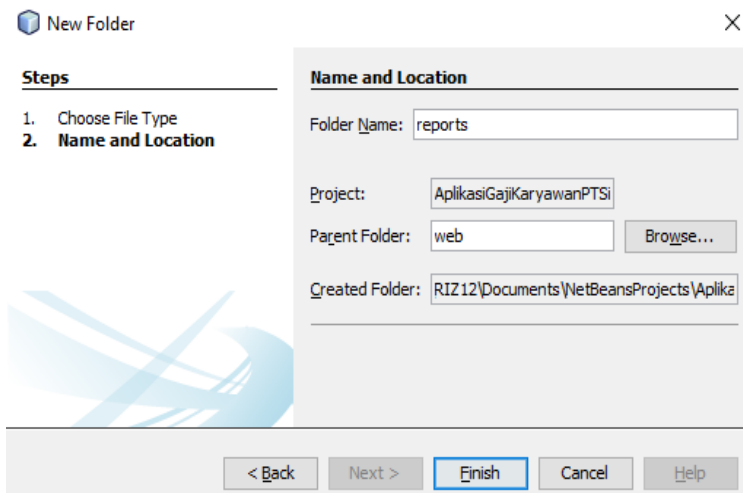
2. Menambahkan Folder Reports

Tambahkan folder Reports di Web Pages, klik kanan pada Web Pages, pilih New > Folder, jika tidak ada pilihan Folder maka pilih Other, lalu pada Categories pilih Other dan File Types pilih Folder.



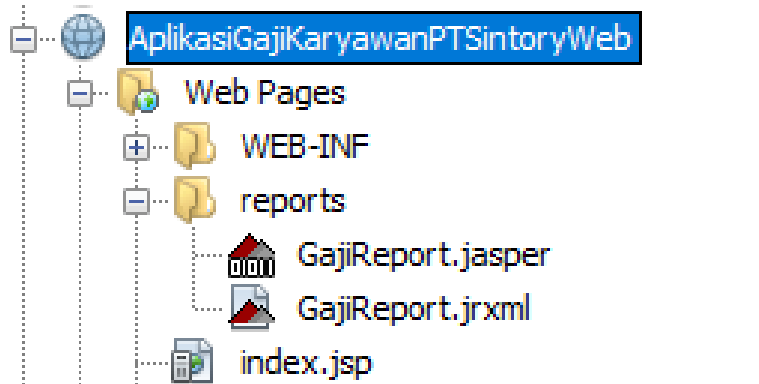
Gambar 291. Membuat Folder Report

Ubah nama folder menjadi reports, klik Finish.



Gambar 292. Mengubah Folder Name

Buka folder reports di dalam folder Source Desktop (file akan diberikan) di materi Copy file GajiReport.jrxml, Paste ke folder reports.



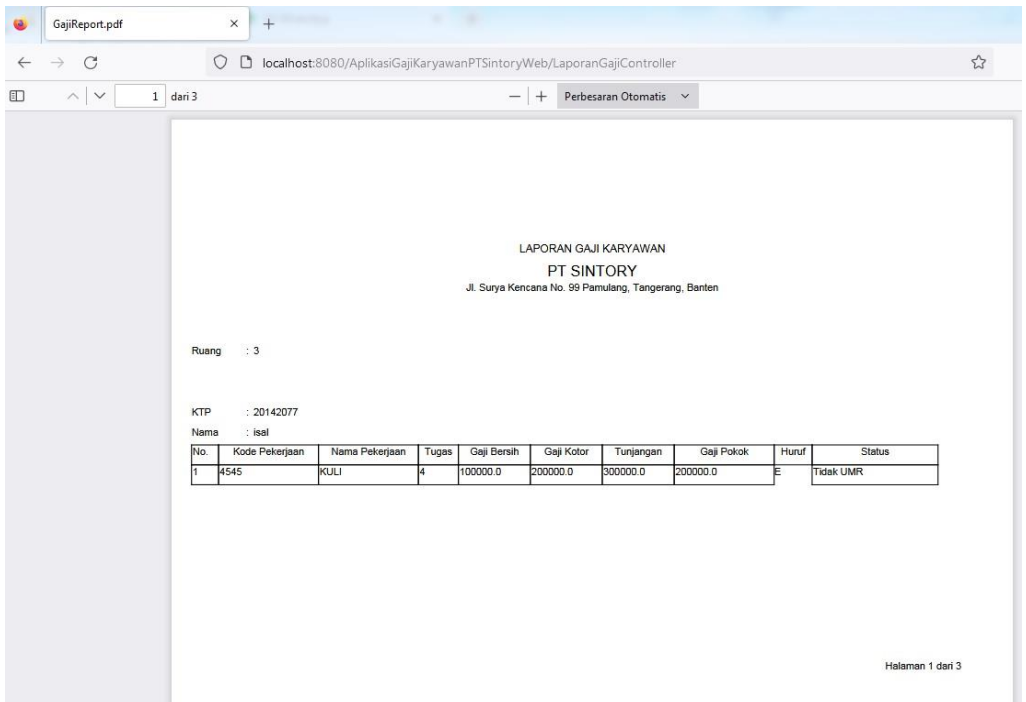
Gambar 293. File Reports

Kemudian Eksekusi (run) aplikasi, coba masuk (login), dan pilih menu Laporan lalu pilih Gaji.



Gambar 294. Menu Laporan

Maka hasilnya akan seperti berikut.



Gambar 295. Hasil Laporan

C. Latihan

Buat report untuk penjualan barang dagang!

- Data stock barang
- Data transaksi penjualan

D. Referensi

- Budiharto, W. (2004). Pemrograman Web Menggunakan J2EE. Jakarta: Elexmedia Komputindo. JENI, T. P. (2007). JENI 1-6.
- Wijono, S. H., Suharto, B. H., & Wijono, M. S. (2006) Pemrograman Java Servlet dan JSP dengan Netbeans. Yogyakarta: C.V ANDI OFFSET.